

Course Introduction

Welcome to Python Programming

Control flow and conditionals

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Practical Examples
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on algorithmic complexity
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control flow and loops to solve a
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with Python
10 questions

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Practicing control flow and loops

Controlling loops

So far, you have only looped over sequences based on the length of the data you wanted to iterate over. In many cases, this is not necessary and depending on the amount of the data it can also be quite costly. You'll now examine how you can control the flow of the loop and exit out when a specific condition is met. You will also look at control statements such as **break**, **continue** and **pass**.

If else

In many cases, you may need to search for a particular item in a list. Once the item is found, there is no need to continue looping over the other results. Using the same example as before, let's assume you only need to check if the dessert "Churros" is in the list and if it is, print a single statement.

```
1 #Starter Code
2 favorites = ['Creme Brulee', 'Apple Pie', 'Churros', 'Tiramisu', 'Chocolate Cake']
3
4 for dessert in favorites:
5     if dessert == 'Churros':
6         print('Yes! One of my favorite desserts is', dessert)
7
```

Run
Reset

Running the above code will output the following:

Yes! One of my favorite desserts is Churros

But what happens if you look for a dessert and that dessert is not on the list? The code above is currently not set up to handle this. Let's look for the dessert "Pudding" which is not on the list, and also add an **else** statement to handle the case of when it's not found. If the dessert is not part of the list, you will print a new statement.

```
1 #Starter Code
2 favorites = ['Creme Brulee', 'Apple Pie', 'Churros', 'Tiramisu', 'Chocolate Cake']
3
4 for dessert in favorites:
5     if dessert == 'Pudding':
6         print('Yes one of my favorite desserts is', dessert)
7     else:
8         print('No sorry, that dessert is not on my list')
9
```

Run
Reset

Running the above code will result in the following output:

No sorry, that dessert is not on my list

Loop control statements

Break

The code works as intended but you may notice one flaw. If you change the search term to something that is on the list like "Churros" and run it again you will get the following output:

Yes one of my favorite desserts is Churros No sorry, not a dessert on my list

This is not what you want! The dessert is on the list but it still printed out the **else** condition. To fix it, you need to use a control statement called **break**.

Add the following:

```
1 #Starter Code
2 favorites = ['Creme Brulee', 'Apple Pie', 'Churros', 'Tiramisu', 'Chocolate Cake']
3
4 for dessert in favorites:
5     if dessert == 'Pudding':
6         print('Yes one of my favorite desserts is', dessert)
7         break
8     else:
9         print('No sorry, not a dessert on my list')
10
```

Run
Reset

Running the above code will resolve the issue. The **break** statement is used to stop the loop, which in turn also stops the **else** condition. Without the **break** the loop will continue even after the **if** condition is satisfied.

Continue

Similar to **break**, **continue** can be used to control the iteration of the loop. The key difference is that it can allow you to skip over a section of the loop but then continue on with the rest. If you change your code to this, you will notice the outcome will print everything except the Churros dessert.

```
1 #Starter Code
2 favorites = ['Creme Brulee', 'Apple Pie', 'Churros', 'Tiramisu', 'Chocolate Cake']
3
4 for dessert in favorites:
5     if dessert == 'Churros':
6         continue
7     print('Other desserts I like are', dessert)
8
```

Run
Reset

Pass

The **pass** statement allows you to run through the loop in its entirety and effectively ignore that the **if** condition has been satisfied.

```
1 #Starter Code
2 favorites = ['Creme Brulee', 'Apple Pie', 'Churros', 'Tiramisu', 'Chocolate Cake']
3
4 for dessert in favorites:
5     if dessert == 'Churros':
6         pass
7     print('Other desserts I like are', dessert)
8
```

Run
Reset

✔ **Completed** **Go to next item**

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